

Busy programs

Computers and programs have become an invisible part of daily life. Every day, people benefit from very complex computer programs that have been written to solve incredibly tough problems.

SEE ALSO

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◀ **192–193** Storing data in files

Compressing files

Almost every type of file that is sent over the Internet is compressed (squeezed) in some way. When a file is compressed, data that isn't needed is identified and thrown away, leaving only the useful information.



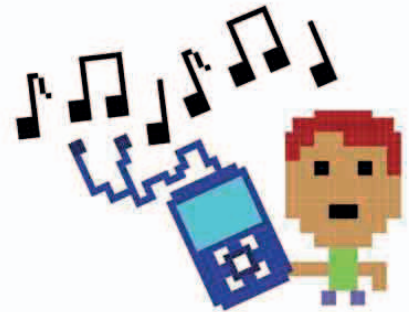
◀ Squeezing data

Compressing a file is like squeezing a jack-in-the-box to make it fit into a smaller space.

REAL WORLD

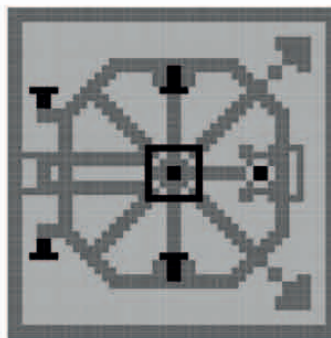
Music files

Without music compression programs, you could only fit a few songs on your music player. By compressing audio files, the average smartphone can now hold thousands of songs.



Secret codes

When you log in to a website, buy something, or send a message across the Internet, smart programs scramble your secret data so that anyone who intercepts it won't be able to understand it. Global banking systems rely on these advanced programs capable of hiding secret information.



◀ Cryptography

Cryptography is the study of codes. Complex mathematical codes scramble and unscramble personal data to keep it safe from thieves.